

Haoran Sun

Department of Economics, Durham University Business School, Durham, DH1 3LB

Email: haoran.sun@durham.ac.uk

Website: haoran-sun.com

Mobile: 07722100722

ORCID: [0000-0001-8805-1942](https://orcid.org/0000-0001-8805-1942)

Research Profile

My research combines game-theoretic modelling of industrial policy and taxation with applied microeconomic analysis of firm productivity, innovation support, and technology adoption. On the theoretical side, I develop models of industrial policy under network externalities and of international taxation of digital businesses. On the empirical side, I use ONS microdata to analyse high-growth business dynamics and identify the firm-level effects of public policy. My published work includes one ABS 4 and two ABS 3 journal articles, alongside two papers under review and three further papers at advanced stages. My forward research programme links theoretical predictions on network externalities to empirical analysis of AI adoption and public procurement, and will underpin future external funding applications, including an ESRC New Investigator Grant.

Job Market Paper

Industrial Policy with Network Externalities: Race to the Bottom vs. Win-Win Outcome

Nigar Hashimzade and Haoran Sun

Studies industrial policy in a two-country model with strategic trade and R&D under network externalities. Shows that policy competition can generate either a race to the bottom or a win-win welfare outcome, depending on the strength of network effects, the degree of product differentiation, and whether innovation is product- or process-based.

Employment

- 2025– Researcher (PDRA), Durham University Business School
Smart & Scale Up project with an external industry partner
Analysing regional scaleup dynamics and their economic impact in North East England using firm-level microdata (Business Structure Database, Longitudinal Small Business Survey) and econometric methods including decomposition analysis and treatment effect estimation. Produced research report for a business support provider in the North East of England that directly informed their advisory service targeting and programme design. Contributing to targeted publications in ABS 3/4 journals.
- 2023–2025 Research Fellow, University of St Andrews
ESRC Funded (Award ES/W010259/1) – Understanding how constraints on access to finance and under-investment impact on productivity growth in smaller firms
Contributed to data management, econometric analysis, and co-authorship across work packages examining SME finance, investment, and productivity using firm-level microdata.

Education

- 2017–2023 PhD in Economics, Durham University
ESRC Funded – Tax Incentives for Innovation
Thesis: [The Economics of Innovation, Investment, and Taxation](#)

2018

2015–2017 Visiting PhD in Economics, University of Exeter
2012–2015 MSc in Economics (Distinction), Durham University
BA in Economics (2:1 Honours), Durham University

Publications

Spatial Myopia, Financial Hubs and the “Golden Triangle”: Tracing the Uneven Landscape of Exogenous Equity Investments in the UK

Ross Brown, Haoran Sun, Augusto Rocha, Marc Cowling
British Journal of Management (ABS 4)

Unpacking the Money-Distance Nexus: Mapping the Spatial Configuration of Exogenous Entrepreneurial Finance across UK Entrepreneurial Ecosystems

Ross Brown, Augusto Rocha, Haoran Sun, Marc Cowling
Small Business Economics (ABS 3)

Why do Owls Fly While Ostriches Flounder? SME Advice, Impacts and Entrepreneurial Outcomes

Ross Brown, Marc Cowling, Weixi Liu, Sabrina Song, Haoran Sun
Journal of Small Business Management (ABS 3)

Papers Under Review

Can We See it in the Numbers Yet? Artificial Intelligence, Firm-Level Productivity and SMEs

Ross Brown, Haoran Sun, Marc Cowling, David Dowell
Journal of Technology Transfer (ABS 3), under review

Does “Slow” Thinking “Speed Up” Performance? The Impact of “Guided Preparation” on Innovation and Productivity in UK SMEs

Ross Brown, Haoran Sun, Weixi Liu, Marc Cowling, Sabrina Song
Journal of Technology Transfer (ABS 3), under review

Working Papers

The Effect of Direct and Indirect Support for Innovation on Productivity: Evidence from Young UK Businesses

John Moffat, Haoran Sun^{*}
Main analysis complete; further empirical work in progress

Taxing the Clouds: An Economic Analysis

Nigar Hashimzade, Haoran Sun^{*}
Manuscript being prepared for submission

Work in Progress

Finance, Growth and Productivity in Family-Owned SMEs

Ross Brown, Marc Cowling, Haoran Sun
ERC–DBT grant awarded (2026); analysis of LSBS data in progress

Do Science Parks Act as a Magnet for Investment? The Role of University Proximity

Ross Brown, Marc Cowling, Haoran Sun
Dataset constructed; empirical analysis forthcoming

Grants

ERC–DBT Small Grant Funding: A Study of Finance, Growth and Productivity in Family-Owned SMEs, £5,000

Ross Brown, Marc Cowling, Haoran Sun (Co-Investigator)

2024 ERC–DBT Small Grant Funding: The Impact of “Guided Preparation” on Innovation and Productivity in UK SMEs, £5,000

Ross Brown, Marc Cowling, Haoran Sun (Co-Investigator)

Policy & Stakeholder Engagement

2025 Presentation of North East High Growth Business report to a major North East business support provider

Findings directly informed advisory targeting and programme design Durham

2025 Smart & Scale Up project kickoff meeting with a major North East business support provider

2025 *Stakeholder partnership briefing* Durham University Business School
ERC–DBT Longitudinal Small Business Survey Dissemination Event

Policy dissemination to government and industry audience London

Academic Presentations

2024 ZEW: Competition and Innovation Workshop

Academic workshop Montenegro

2024 Research Review, ESRC Team Project Meeting (Award ES/W010259/1)

Funder review Oxford Brookes University

2024 “Unpacking the Money-Distance Nexus,” Internal Research Seminar

Research seminar University of St Andrews

Teaching

2026 Guest Lecturer – *What Shaped This Data? Platform Economics and the Limits of Analytics*

MSc Digital Marketing and MSc Digital Marketing Strategy Newcastle University

2020 Teaching Assistant – Macroeconomics (UG)

Led seminar groups of up to 20 students covering economic growth, inequality, microfoundations, and intergenerational macroeconomics; facilitated presentations, debates, and problem sets; marked coursework and provided individual feedback Durham University

2019 Teaching Assistant – Corporate Finance (UG)

Marked coursework and provided individual feedback Durham University

Teaching competencies: Microeconomics, Macroeconomics, Econometrics, Microeconometrics, Public Economics (UG and PG).

Honours & Professional Activities

2017–2022 NINE Doctoral Training Partnership Studentship, ESRC

2017–2021 Research Training Support Grant, ESRC

Technical Skills & Languages

Software: Stata, Python, R, $\text{\LaTeX}$, Maple, Matlab, EViews, SPSS, NVivo

English: Professional proficiency

Mandarin: Native proficiency

References

References available on request.